

Sharon Tey

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EDUCATION

University of California, San Diego

Bachelor's Degree in Data Science

GPA: 3.83/4.00

Sep 2024 - Apr 2027 (expected)

De Anza College

Associate's Degree for Transfer in Computer Science

GPA: 3.96/4.00

Sep 2022 - Jun 2024

SKILLS & INTERESTS

- **Programming:** Python (Pandas, Numpy, Scikit-learn), SQL, C++, HTML, CSS, JS
 - **Data Analysis:** Hypothesis Testing, A/B Testing, Regression and Classification, Time Series Analysis
 - **Data Visualization:** Matplotlib, Seaborn, Tableau
 - **Tools:** Git, Jupyter Notebook, VS Code, Google Workspace (Docs, Sheets, Slides)
 - **Languages:** English (Native), Mandarin (Native), Malay (Proficient)
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PROFESSIONAL EXPERIENCES

Power Transformation Lab, UC San Diego

Dec 2025 - Present

Research Analyst

- **Collaborated with Prof. Michael Davidson** on analyzing China's mid-century energy system transition by synthesizing research findings and preparing analytical report and presentation materials.
- **Streamlined data processing workflows** by consolidating and harmonizing datasets from **6 large-scale energy system models**, enabling consistent cross-model analysis of electricity transition scenarios.
- Built **reproducible Python data pipelines** to automate cleaning, processing, and visualization of multi-model datasets spanning multiple scenarios and decades.
- Transformed complex energy modeling data into **actionable insights through comparative analyses** and visualizations to support research decisions and project discussions.

Resilience Inc.

Jun 2025 - Sep 2025

SEO Analyst Intern

- Conducted **quantitative analysis of website traffic, user engagement, and keyword performance** using Google Analytics to identify optimization opportunities.
 - Collaborated with **cross-functional teams (marketing, content, and product)** to translate analytical insights into strategic initiatives.
 - Developed **standardized SEO process documentation and workflow guidelines**, improving cross-team consistency and reducing onboarding time by 30%.
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PROJECTS

Wearable Assistive Audio Classification System

Jan 2026 - Mar 2026

- Designed and deployed an end-to-end TinyML pipeline on an ESP32 microcontroller, processing live microphone audio through custom C++ MFCC extraction and a TFLite Micro CNN model — no Wi-Fi or cloud dependency required
- Trained a 3-block Conv1D neural network (95K parameters), achieving ~81% accuracy across 5 sound classes, using advanced augmentation techniques including Mixup, SpecAugment, pitch shifting, and time stretching to address class imbalance